

MATH 126 PROBLEM SET 3

Due: Sep 24, 11:59pm

1. This exercise computes the area and volume of the n -dimensional sphere from the Gaussian integral.

(1) Let's first compute the Gaussian integral. Let

$$I := \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Note that

$$I^2 = \left(\int_{-\infty}^{\infty} e^{-x_1^2} dx_1 \right) \left(\int_{-\infty}^{\infty} e^{-x_2^2} dx_2 \right) = \int_{\mathbb{R}^2} e^{-(x_1^2+x_2^2)} dx_1 dx_2$$

Evaluate the integral by changing variables to polar coordinates. (You should find $I = \sqrt{\pi}$.)

(2) Compute

$$I^n = \int_{\mathbb{R}^n} e^{-(x_1^2+x_2^2+\dots+x_n^2)} dx$$

by spherical integration and find that

$$I^n = \text{Area}(\mathbb{S}^{n-1}) \int_0^\infty e^{-r^2} r^{n-1} dr.$$

(3) The Gamma function is defined by

$$\Gamma(z) := \int_0^\infty t^{z-1} e^{-t} dt.$$

Show that $\Gamma(z+1) = z\Gamma(z)$ and that $\Gamma(1) = 1$. Conclude that $\Gamma(n) = (n-1)!$ for any positive integer n .

(4) Write the formula for the area of the unit sphere $\mathbb{S}^{n-1}$ and the volume of the n -dimensional unit ball $B(0,1) \subset \mathbb{R}^n$ in terms of the Gamma function.

2. Show that

$$\frac{\partial}{\partial r} \int_{B(0,r)} u(x) dx = \int_{\partial B(0,r)} u(x) dS(x).$$

3. Consider the wave equation in the rectangle $\Omega = (0,1)^2$

$$\begin{cases} \partial_t^2 u - \partial_x^2 u = 0, \\ u|_{\partial\Omega} = 0. \end{cases} \quad (1)$$

Note that $u = 0$ satisfies the equation. Construct a smooth nonzero solution to the equation. (**Hint:** Recall that by changing to the characteristic coordinates, a solution

takes the general form $f(x+t) + g(x-t)$. Now apply boundary conditions and see what restrictions this imposes.)

Note that this exercise shows that the Dirichlet problem for the wave equation is ill-posed.

4. Borthwick §4.8 exercises 4.2 and 4.3.